

Stopping Is Not Resolving: Why Reasoning Models Loop When No Answer Fits

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Abstract

Reasoning models sometimes never stop thinking, and common fixes simply force them to. We ask why the loops arise and whether stopping them helps. Removing the valid answer from a multiple-choice question makes DeepSeek-R1-Distill-Qwen-7B loop on 75% of problems it otherwise solves (vs. 4–5%), and deleting the correct option from real AQuA questions does the same (74% vs. 9%); most natural loops show the same conflict between the model’s answer and the options. The decision to emit `</think>` is written by the same late MLPs in controlled and natural loops, yet this gate is generic: it is as closed at healthy self-doubt as in loops, and loops never come close to opening it. Opening it, or forcing `</think>`, ends the thinking but not the doubt: deliberation continues after `</think>`, and accuracy is recovered only when the answer being doubted was already right. The model even encodes whether that answer is right (a linear probe reaches AUC 0.77 inside loops), yet stopping where the probe is confident does not make it commit to the answer: the doubt survives the stop.

1 Introduction

Large reasoning models (LRMs) think before they answer, and occasionally they do not stop: the trace runs to the token budget, re-checking the same step, and no answer is produced (Duan et al., 2026). Practical remedies end the thinking from outside, by forcing the end-of-thinking delimiter (budget forcing; Muennighoff et al., 2025) or by steering the model out of the loop (Yu et al., 2026; Zhang et al., 2026), and are judged on whether the loop is broken. We ask what triggers the loop and whether breaking it resolves whatever caused it.

Our contributions: (1) a controlled trigger: removing the valid option makes the model loop on three quarters of multiple-choice questions (MCQs) it can solve, on constructed and on real

questions, and the same answer–option conflict underlies most natural MCQ loops (§3); (2) a mechanism: the `</think>` decision is written by the same late MLPs in controlled and natural loops, but this gate is generic rather than suppressed in loops (§4); (3) stopping is not resolving: every way of opening the gate ends the thinking but leaves the doubt, and helps only when the doubted answer was right (§5); (4) the model’s hidden state encodes whether its committed answer is right, but a probe-guided stop does not turn that knowledge into correct answers (§6).

2 Setup

Model and decoding. DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025), served with vLLM (Kwon et al., 2023), sampled with the recommended $T=0.6$, top- p 0.95, 4 seeds and a 12k-token budget. A **loop** is a run that exhausts the budget without emitting `</think>`. Greedy decoding inflates looping 2–4 $\times$ wherever a valid answer exists (Table 1).

Controlled conditions. 150 MATH (Hendrycks et al., 2021) problems (level ≥ 3 , numeric answers) as two-option MCQs: ALIGNED (correct number vs. an unrelated sentence), SWAP (reversed) and CATCH (a wrong number vs. a sentence: no valid option), plus the open-ended question. To avoid an artificial distractor, we also pose 227 AQuA questions with the correct option deleted (four real wrong options), next to their original and open-ended forms.

Natural benchmarks. The datasets of Yu et al. (2026) in their native form: GSM8K (Cobbe et al., 2021), MATH-500, AQuA (Ling et al., 2017) (5 options), LogiQA (Liu et al., 2020) (4 options).

Pairs and positions. To locate the mechanism we pair a run that loops with one that finishes: CATCH vs. ALIGNED on the same problem and

	valid answer?	sampled	greedy
open-ended	yes	4.2	9.3
ALIGNED (A correct)	yes	5.3	11.3
SWAP (B correct)	yes	4.8	8.0
CATCH (A wrong)	no	74.7	75.3
AQuA, open-ended	yes	2.5	–
AQuA, original 5 options	yes	8.5	–
AQuA, correct option removed	no	74.4	–
GSM8K (open)	yes	–	4.2
MATH-500 (open)	yes	5.2	10.0
AQuA, all 254 (5 options)	yes	7.9	25.6
LogiQA (4 options)	yes	9.5	39.4

Table 1: Loop rate (%): the run never emits `</think>` within 12k tokens. Sampled = mean over 4 seeds (range $\leq \pm 4.7$). Top: 150 MATH problems; middle: the same 227 AQuA questions in three forms; bottom: natural benchmarks.

seed, CATCH looping in one seed and finishing in another, and natural questions that loop in one seed and finish in another (identical prompt); 426 pairs (272 controlled, 154 natural). We compare the token before `</think>` in the finished run with the looping run’s *commit point*, the token after which, having just stated an answer, it writes “Wait” (detected heuristically, hand-checked; Appendix).

3 When do reasoning models loop?

Removing the valid answer makes the model loop on 74.7% of runs on problems it solves in every other form (90–95% accuracy), a $15\times$ increase that is unchanged under greedy decoding (Table 1). The traces are explicit: the model derives the answer, notes it is not an option, re-derives it, and repeats. The effect does not depend on the artificial distractor: deleting the correct option from real AQuA questions gives 74.4%, against 8.5% with the five options and 2.5% open-ended. Runs without a valid option that do finish pick a wrong option (82% for CATCH, 100% for AQuA) and never decline, the behaviour G3ral et al. (2025) report for instruction-tuned models.

The conflict also arises naturally: MCQ benchmarks loop more than open-ended ones (8–10% vs. 5%), and 23 of 30 AQuA questions that loop in one seed but finish in another say, in the loop, that the model’s answer is not among the options. Here the conflict comes from the model’s own error (or from noisy answer keys).

	AQuA	LogiQA	MATH	CATCH
left looping	0	0	0	–
resampled from commit (4k)	7	20	30	–
restore MLPs	12	31	31	–
force <code></think></code>	14	34	35	–
boost logit(<code></think></code>)	24	36	39	–
<i>doubted answer was correct</i>	42–58	59–66	48–58	–
<i>doubted answer was wrong</i>	6–18	14–18	13–18	–
answer runs to the 1.2k cap	45	31	63	72
finished-run ceiling	50	71	90	–

Table 2: Interventions at the commit point of natural loops (accuracy, %; 3 sampling seeds per pair). 95% bootstrap CIs over pairs are ± 8 –15 points (e.g. MATH force 22–51, restore 18–44); the three stops do not differ. All three end 89–100% of loops at once; random, `</think>`-direction-only and MLP-15 edits stop 0–7%. Split rows: range over the three stops ($n = 4/36/16$ correct, 24/59/15 wrong).

4 Where is the stop decided?

At a real stop `</think>` has logit $\approx +32$ and is the top-1 token in every pair; at the loop’s commit point it is $\approx +5$ (gap > 0 in 100% of pairs in every group, Wilcoxon $p \leq 4 \times 10^{-6}$). Direct logit attribution (nostalgebraist, 2020; Nanda and Bloom, 2022) assigns 90–92% of the gap to MLPs in layers 20–27, with the same profile in controlled and natural pairs (Spearman $\rho = 0.98$ –1.00; Figure 1b). The gate is causal: replacing these MLPs’ output at the commit point with their output at a real stop recovers 76–80% of the gap, a mean vector from the controlled pairs recovers 79–92% in natural loops, and the same patch on MLP 15 recovers $\approx 0\%$.

The gate is generic. If loops suppressed the gate, the stop signal should be lower at a loop’s commit point than at a healthy doubt. It is not (Figure 1a): at the same position in the trace, loop doubts are +0.35 logits *higher* (95% CI 0.08–0.62; regression on 3.4k doubt points with log-position and dataset), against $\approx +26$ for a real stop. Without any pairing, across all 460 natural loops the highest $P(\text{code}</think>)$ at any line break is ≈ 0.001 (median), and that closest point is where the stop MLPs push hardest (+9 to +12 vs. +5 at a typical line break and +24 at a real stop). The late MLPs are the model’s general “stop now” gate; a loop is a run that never reaches the state that opens it.

5 Does stopping fix loops?

We open the gate at each natural loop’s commit point by restoring the MLPs once, forcing

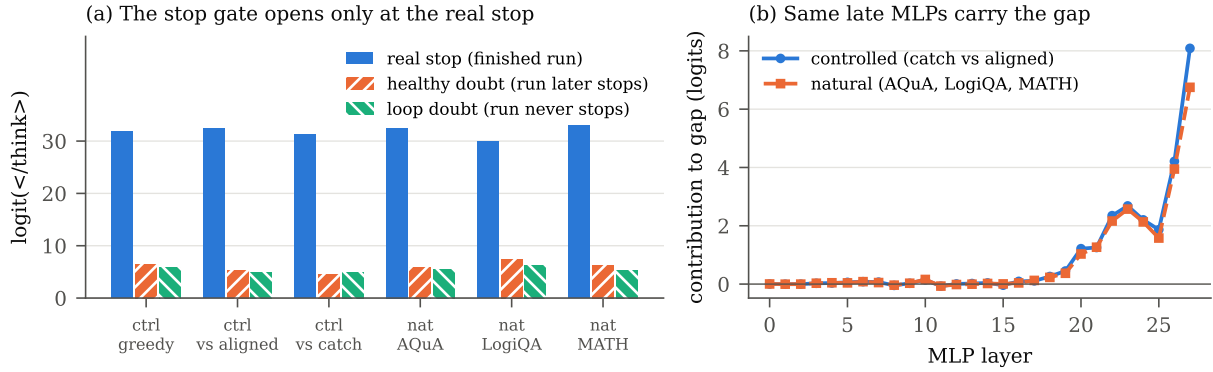


Figure 1: (a) Logit of `</think>` at the real stop of finished runs, at healthy doubts in finished runs (an answer is stated, then “Wait”, but the run later stops) and at the commit point of looping runs. (b) Direct contribution of each MLP to the gap between the real stop and the loop’s commit point, for controlled and natural pairs.

`</think>`, or adding the matched logit boost (Table 2). All end the thinking at once; controls do not. Stopping costs nothing relative to letting the loop run 4k more tokens, but the three ways of stopping are equivalent and far below the ceiling, for two reasons. First, the doubt survives: the answer often opens with “But wait” and keeps re-checking (median 11–13 “Wait”s) until the answer budget runs out (31–63% of natural, 72% of CATCH loops). Second, stopping freezes the doubted answer: when it was right, stopping returns it 42–66% of the time against 6–39% left alone; when it was wrong, accuracy stays at 6–18%. Most AQuA loops (24/28) doubt a wrong answer. The same holds without pairs: applying the controlled MLP donor at the model’s own closest-to-stop point in 60 natural loops stops all of them, with 33% correct (forcing `</think>`: 30%; MLP-15 control: 17% stop, 8% correct).

6 The model knows, but stopping does not help it act

Stopping helps exactly when the doubted answer is correct, but the text does not reveal when: how often a loop re-states its answer does not predict that it is right (31–40% at every count). The hidden state does. We train a logistic-regression probe on the layer-24 residual at 6.1k “states an answer, then doubts it” points (and real stops) from 1.6k sampled runs, labelled by the answer key, with 5-fold splits by question. It predicts correctness with AUC 0.88 overall and 0.77 at loop points (position baselines: 0.51); within datasets, 0.87 (MATH-500) and 0.80 (AQuA), but 0.53 (chance) on LogiQA, so the signal is specific to mathematical answers, in line with Zhang et al. (2025).

We then evaluate it as an online stopping rule on the 460 recorded natural loops: replaying each loop and using only the text so far, it stops at the first commit point where the held-out probe gives $P(\text{correct}) > 0.7$, and the answer is then generated. The probe picks better points by the stated answer (40% correct vs. 21% at the first commit point), but not by the answer the model then writes: 19–36% correct for AQuA, LogiQA and MATH-500 (MLP switch or forced `</think>`), no better than stopping blindly, and because it stops only 40–49% of these loops, its overall accuracy (8–18%) is lower than stopping every loop (14–28%). Even when the model’s state encodes that its answer is right, a forced stop does not make it commit to that answer.

7 Related work

Loops in LRMs have been characterised through attention (Duan et al., 2026; Zhang et al., 2026) and broken by steering (Yu et al., 2026) or budget forcing (Muennighoff et al., 2025); we isolate a trigger and show that breaking a loop does not resolve it. Missing premises cause verbose reasoning (Fan et al., 2025), and instruction-tuned models pick invalid options (Góral et al., 2025); a reasoning model facing the same conflict fails to terminate. Hidden states encode answer correctness (Zhang et al., 2025); we show that acting on this signal with a forced stop does not recover it.

8 Conclusion

Reasoning models loop when their answer cannot be reconciled with the options, a condition we create on constructed and real questions and that natural MCQ loops reproduce through the model’s

own errors. A generic late-MLP gate decides when to stop, and loops never open it. Forcing it open ends the thinking but not the doubt, so loop-breaking methods should be judged on what happens after the stop. The model even carries the missing signal, whether its answer is right, but a stop does not let it use it: resolving loops will require acting on the conflict itself, not on the stop.

Limitations

We study one model ([TODO: second model]); results may differ for larger or differently trained LRMs. Commit points come from a heuristic detector (hand-checked: 11/12 controlled, 9/10 AQuA/LogiQA, 19/31 MATH-500, all 13 MATH pairs that commit to the gold answer correct); the comparison of a real stop with a doubt is partly built in by selection, which is why we rely on the doubt-vs-doubt and position-matched comparisons. Interventions use three samples per pair; AQuA contributes 28 pairs. The probe’s labels are the correctness of the stated answer, not of the answer written after a forced stop, and it fails on LogiQA; the stopping rule is evaluated by replaying recorded loops, not in live generation, and only on runs known to loop, so its effect on runs that would have finished is untested. Activations are bfloat16, to which late layers are sensitive. We do not identify the upstream computation that opens the gate, and loops are defined by a 12k-token budget.

References

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

DeepSeek-AI. 2025. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*.

Zenghao Duan et al. 2026. Circular reasoning: Understanding self-reinforcing loops in large reasoning models. *arXiv preprint arXiv:2601.05693*.

Chenrui Fan, Ming Li, Lichao Sun, and Tianyi Zhou. 2025. Missing premise exacerbates overthinking: Are reasoning models losing critical thinking skill? In *Conference on Language Modeling (COLM)*. ArXiv:2504.06514.

Gracjan Góral, Emilia Wiśnios, Piotr Sankowski, and Paweł Budzianowski. 2025. Wait, that’s not an

option: LLMs robustness with incorrect multiple-choice options. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*. ArXiv:2409.00113.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the MATH dataset. In *NeurIPS Datasets and Benchmarks*.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*.

Wang Ling, Dani Yogatama, Chris Dyer, and Phil Blunsom. 2017. Program induction by rationale generation: Learning to solve and explain algebraic word problems. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*.

Jian Liu, Leyang Cui, Hanmeng Liu, Dandan Huang, Yile Wang, and Yue Zhang. 2020. LogiQA: A challenge dataset for machine reading comprehension with logical reasoning. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI)*.

Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori Hashimoto. 2025. s1: Simple test-time scaling. *arXiv preprint arXiv:2501.19393*.

Neel Nanda and Joseph Bloom. 2022. TransformerLens. <https://github.com/TransformerLensOrg/TransformerLens>.

nostalgebraist. 2020. interpreting GPT: the logit lens. LessWrong.

Sheldon Yu, Tong Yu, Xunyi Jiang, Rohan Surana, Gagan Mundada, Sungchul Kim, Lina Yao, Julian McAuley, and Junda Wu. 2026. Can we break LLMs out of self-loops? fine-grained reasoning control with activation steering. *arXiv preprint arXiv:2607.18100*.

Anqi Zhang, Yulin Chen, Jane Pan, Chen Zhao, Aurajit Panda, Jinyang Li, and He He. 2025. Reasoning models know when they’re right: Probing hidden states for self-verification. *arXiv preprint arXiv:2504.05419*.

Yuanhe Zhang, Ziwei Wang, Jie Ren, Haoran Gao, Zhenhong Zhou, Fanyu Meng, Cong Wu, Li Sun, and Sen Su. 2026. When reasoning goes astray: Attention dynamics of uncontrolled reasoning. *arXiv preprint arXiv:2609.38817*.

A Implementation details

Mechanism analyses use TransformerLens with RMSNorm folding and unembedding centering applied in place on the GPU (matching an unprocessed model to bf16 precision), KV-cached batched decoding, and positions mapped through tokenizer character offsets. A finished run is rebuilt as reasoning + `</think>` + answer exactly as generated: inserting an extra newline before `</think>` moves the “stop” to a position where $P(</think>) \approx 0$, an error we found in and corrected for an earlier analysis. MATH answers are scored with math-verify. Code and data will be released.